

LECTURE 2 — ENERGY GENERATION

PHY199-RAP · Energy in the 21st Century

Dr. Salvatore Rappoccio

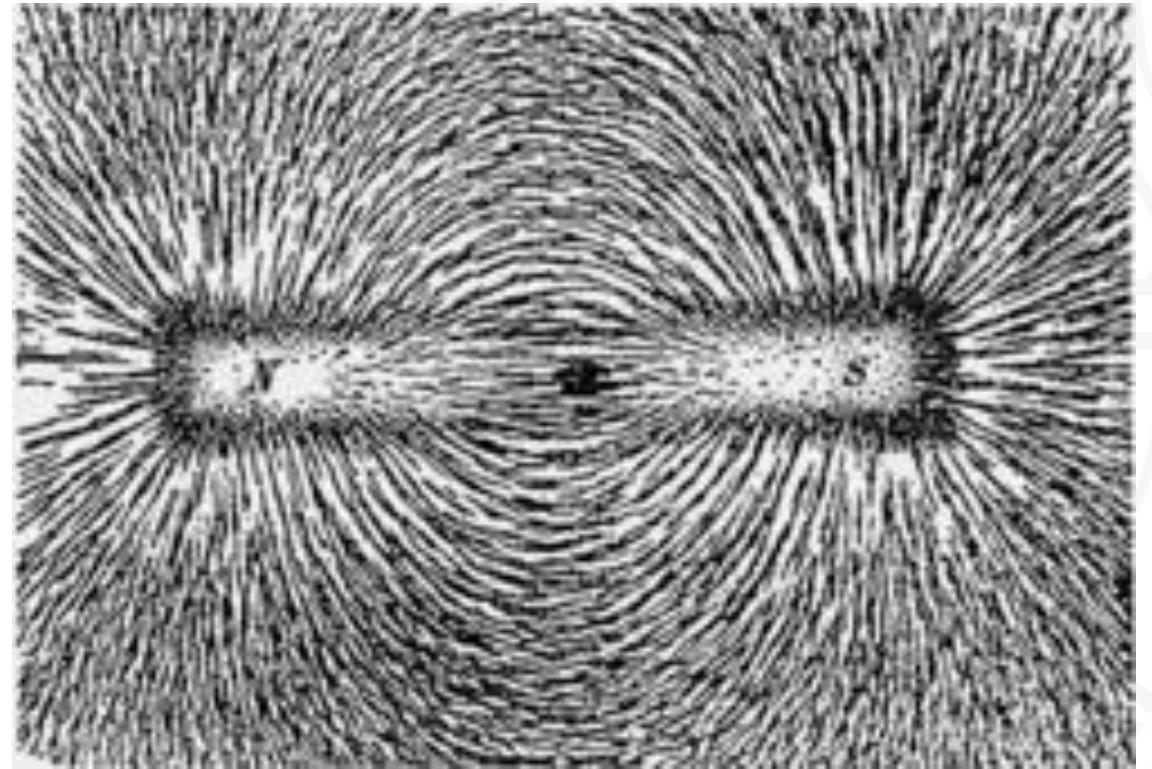


MAGNETISM



Magnets

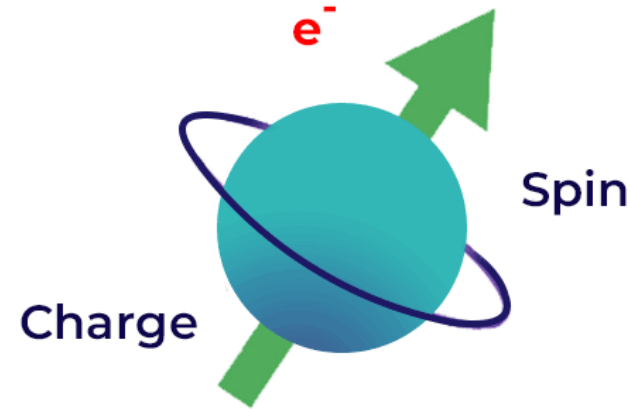
- Magnets have been known to humans for centuries
- Similar to electricity, there are two poles (North and South)
- Distinctly, however, they ALWAYS come in pairs
 - You can have a single positive charge or a single negative charge, but there is no such thing (that we know of) of a magnetic “monopole”



<https://en.wikipedia.org/wiki/Magnet>

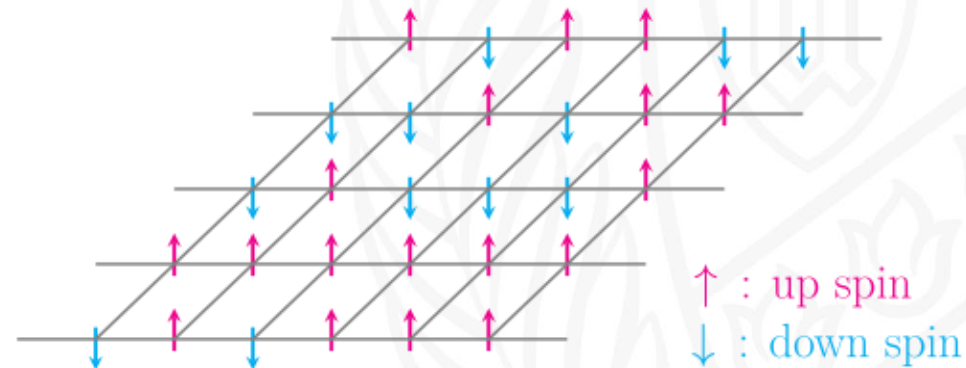
Magnets

- The way permanent magnets work is ultimately quantum mechanical
- Electrons and protons have a quantum property that acts like a “spin”
- However, it isn’t actually spinning!
 - As far as we know, electrons are pointlike (it has a radius less than 10^{-18} m), so can’t spin
- Spins want to be aligned if they can, so the spins at the microscopic level add together to make a macroscopic spin



<https://www.geeksforgeeks.org/physics/spintronics/>

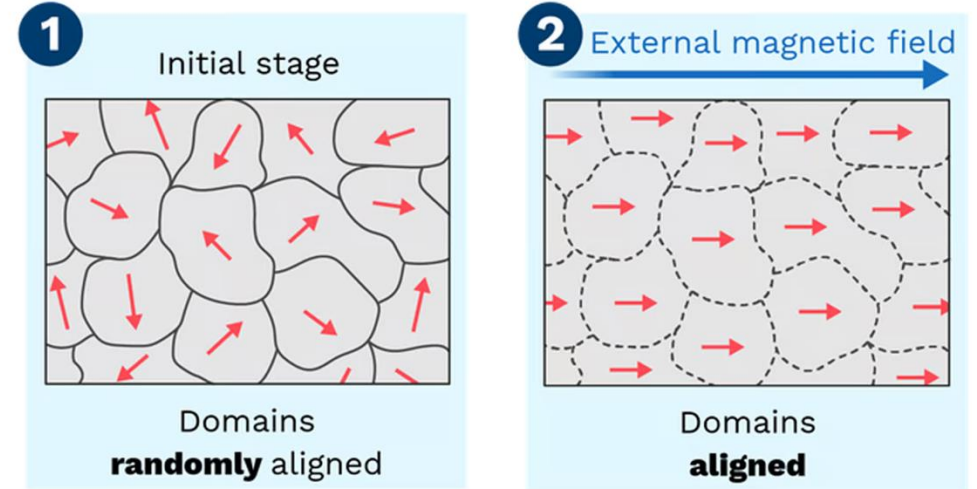
This is a lie, it is not actually spinning, it just ACTS like it's spinning... sort of



https://en.wikipedia.org/wiki/Ising_model

Magnets

- Macroscopically, certain metals (like iron) can have DOMAINS of spin
- If you put them into another magnetic field, they will align and retain their magnetization



<https://sdmmagnets.com/orientation-directions/>

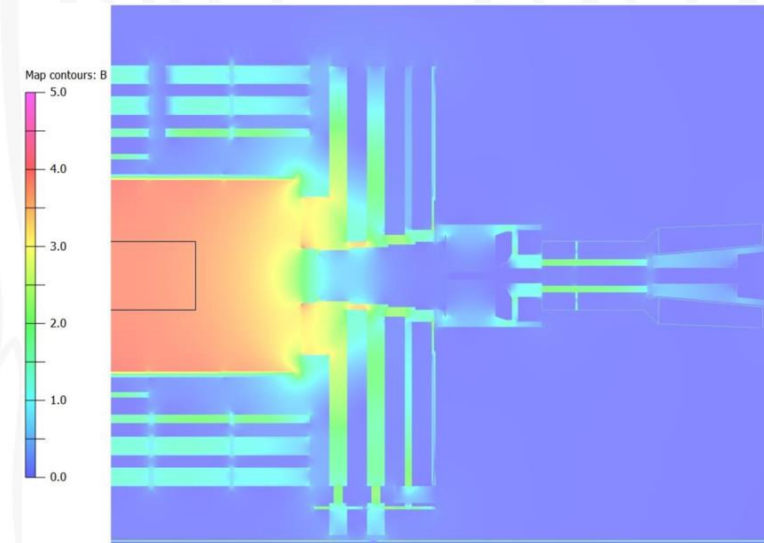
Magnets

You ARE responsible to understand Teslas as a unit of magnetic field

- Strength of magnets is measured in Teslas:

$$1 \text{ Tesla} = 1 T = \frac{1 N}{C \cdot m/s} = \frac{1 N}{A \cdot m}$$

- 1 Tesla is a HUGE magnetic field. The experiment I work on (Compact Muon Solenoid) has a 3.8 Tesla magnetic field, and it is the largest flux magnet on earth.



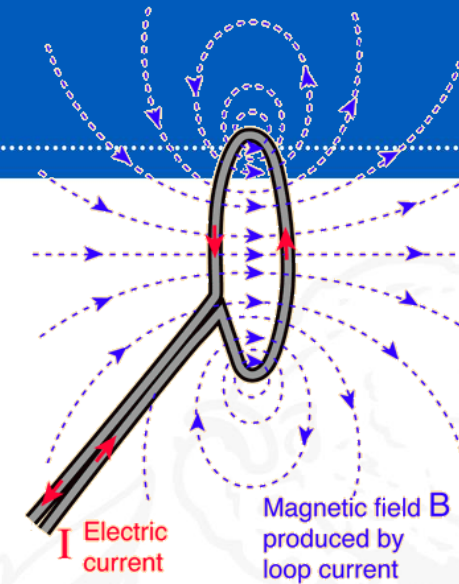
Magnets from Electricity

- If you have a current, the current INDUCES a magnetic field
- Wrap a wire around an iron core and BOOM. Magnet.



Magnetic Fields

- Magnets have a magnetic field just like charges have an electric field
- **Circular loop carrying current i :**



Current:
Ampere = Coulombs / second

Magnetic field:
1 Tesla = 1 Newton / Amp-meter

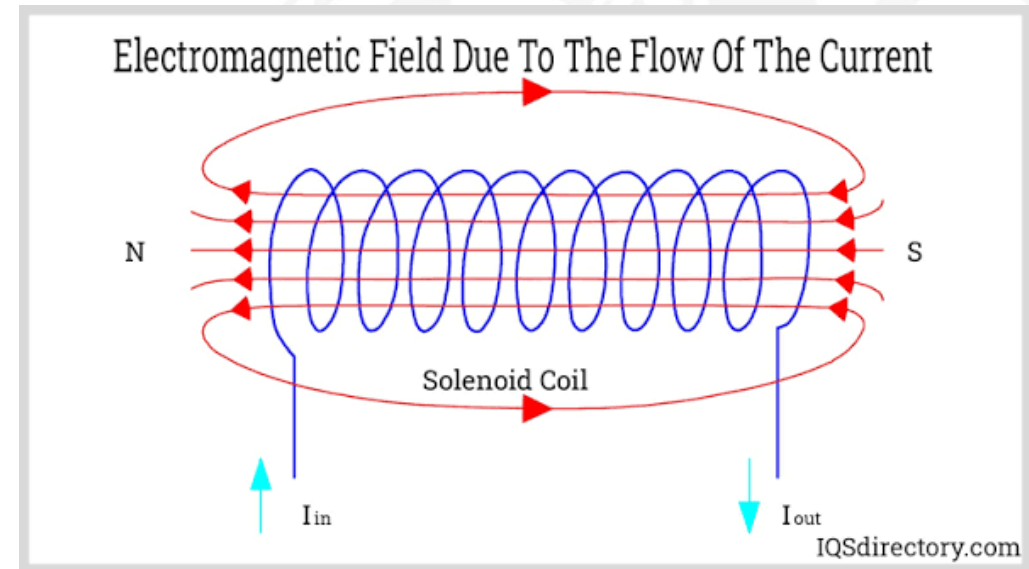
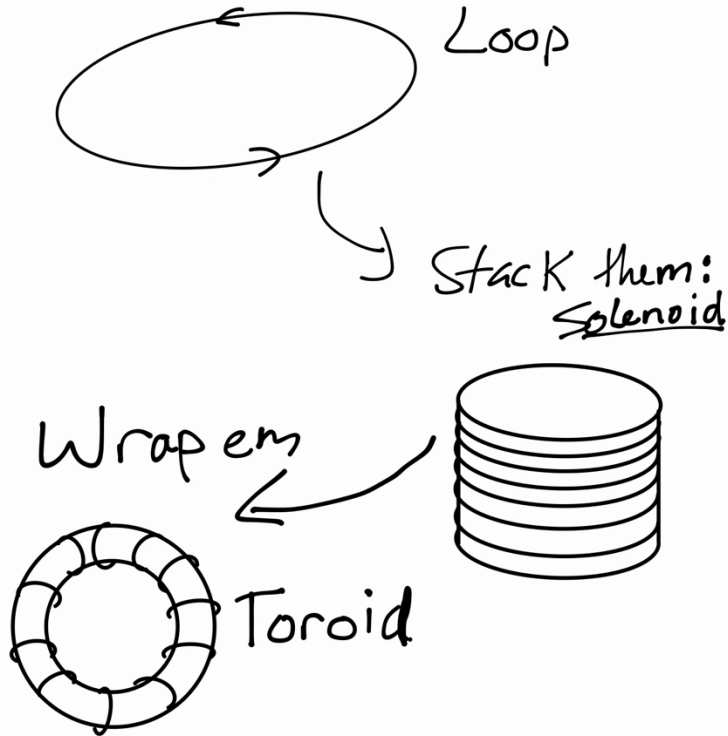
$$B = \frac{\mu_0 i}{2R}$$

Permeability of free space (constant):
 $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$

Radius of loop:
meters

Magnetic Fields

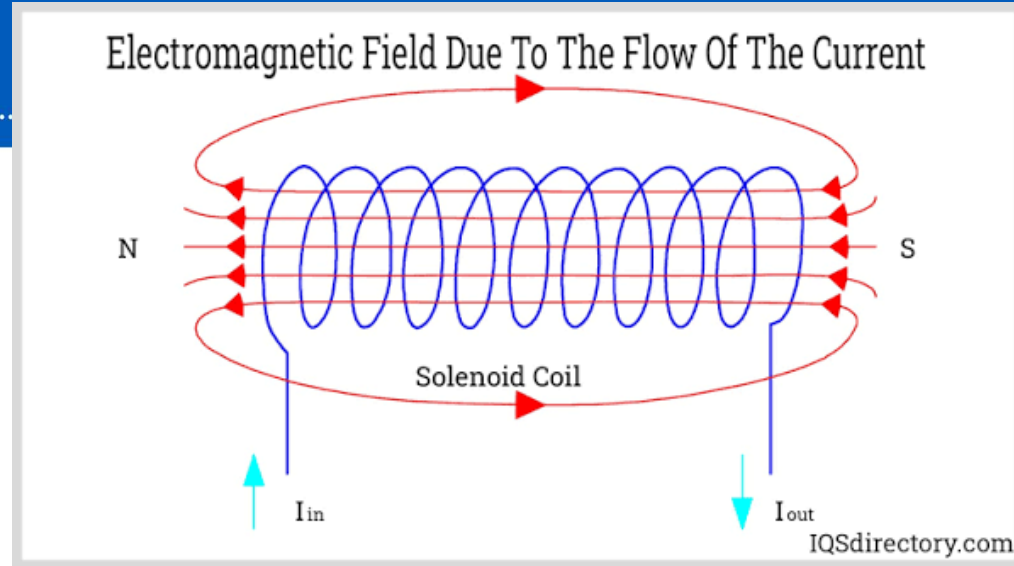
- When you stack loops together, the magnetic fields add to form a **solenoid**:



<https://www.iqsdirectory.com/articles/electric-coil/solenoid-coils.html>

Magnetic Fields

- We won't derive the magnetic field of a solenoid (Ampere's law, PHY108)
- Just quote the result:



Magnetic field:
1 Tesla = 1 Newton / Amp-meter

$$B = \frac{\mu_0 i N}{L}$$

Number of turns

Length of solenoid

Permeability of free space
(constant):

$$\mu_0 = 4\pi \times 10^{-7} T \cdot m/A$$

You ARE responsible to understand the magnetic field of a solenoid is proportional to the current and the number of turns per unit length

Magnetic Flux

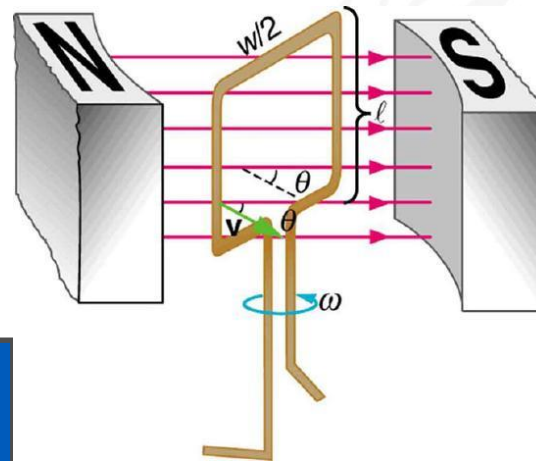
- An important concept we're going to use is the idea of a magnetic flux
- For simple cases (constant magnetic field passing directly through an area) it is the product of the magnetic field times the area:

Magnetic flux:
1 Weber = 1 Tesla · Amp

$$\Phi_B = B \cdot A$$

Magnetic field:
1 Tesla = 1 Newton / Amp-meter

Area: m²



For those who love calculus like me:

The flux is the integral of the dot product of the magnetic field and the plane of the area.

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

You ARE responsible to understand the flux of a magnetic field is the product of the magnetic field and the area

ELECTRICITY AND MAGNETISM : NOW TOGETHER!

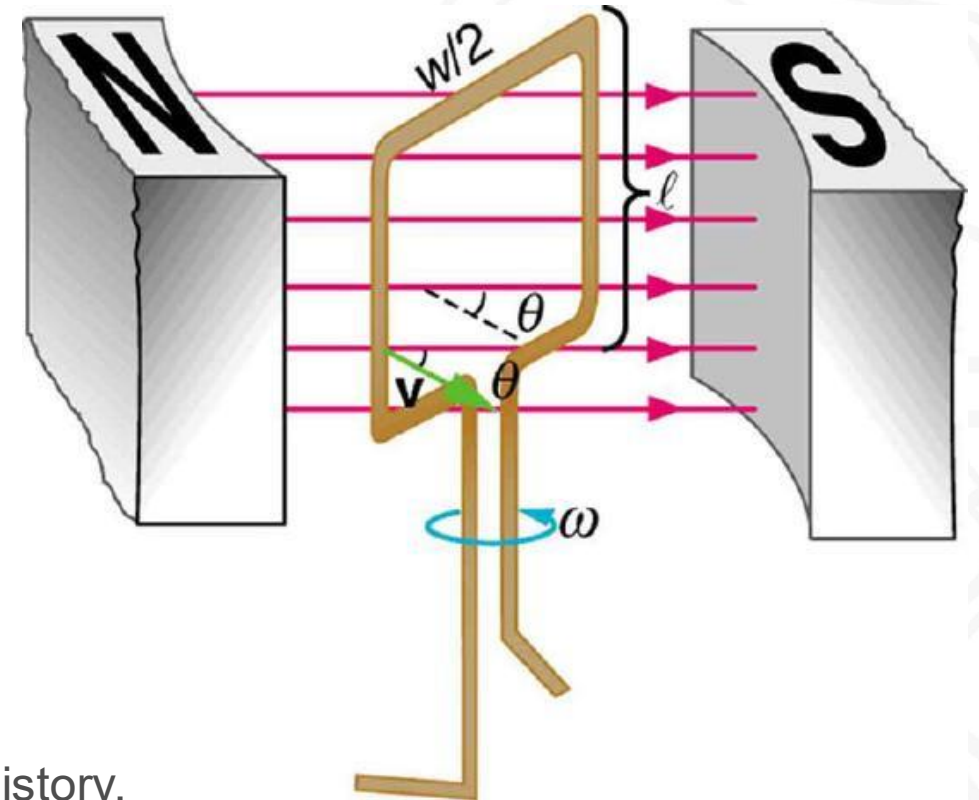


Magnetic Induction

- The modern electrical grid is built upon the [law of induction](#).

**A current can be created
("induced") by a changing
magnetic flux**

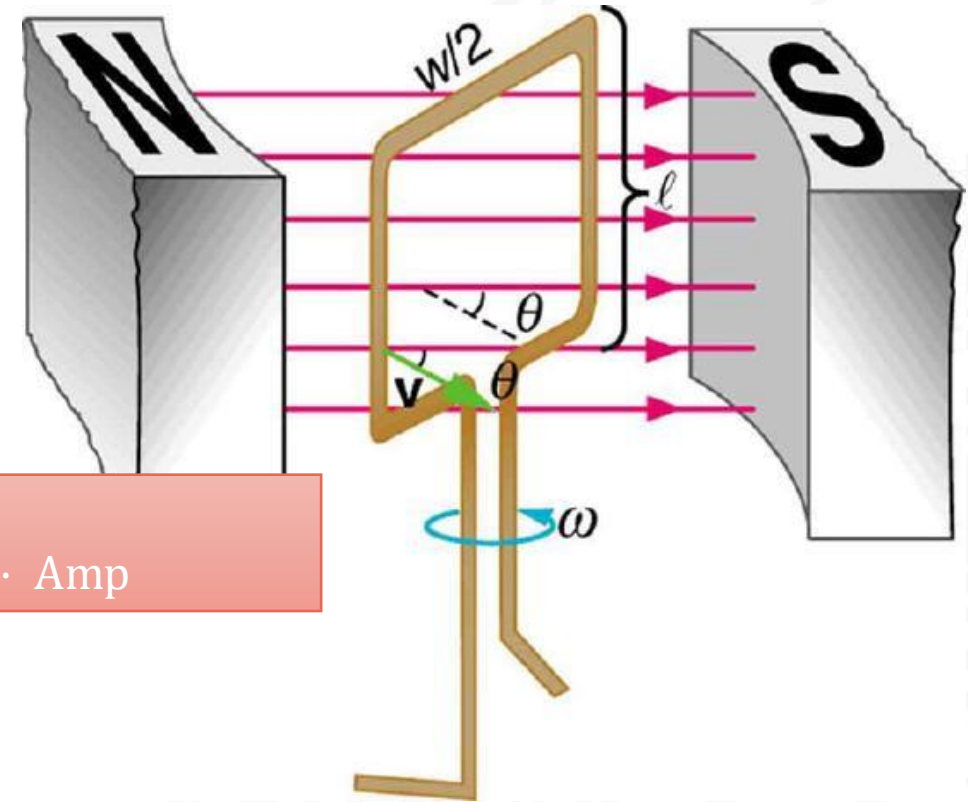
Arguably the most influential scientific statement in human history.
This allows us to generate electric power.



You absolutely need to understand this. Every human should.

Magnetic Induction

- Mathematically:
 - Electromotive force (voltage) is generated by changing magnetic field



Electromotive force (Voltage):
1 Volt

Magnetic flux:
1 Weber = 1 Tesla · Amp

$$\mathcal{E} = \frac{\Delta\Phi_B}{\Delta t}$$

Time:
seconds

You ARE responsible to understand a voltage is induced from a changing magnetic flux

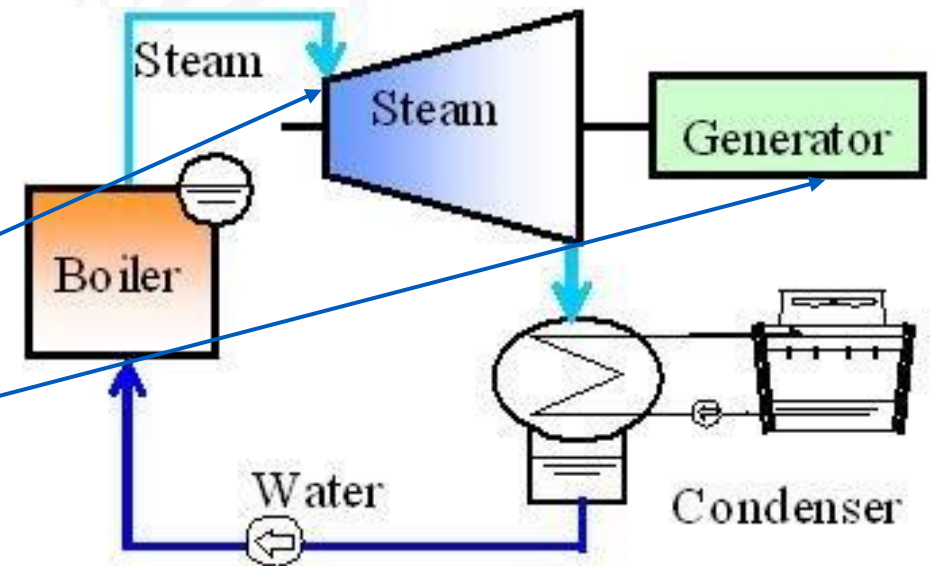
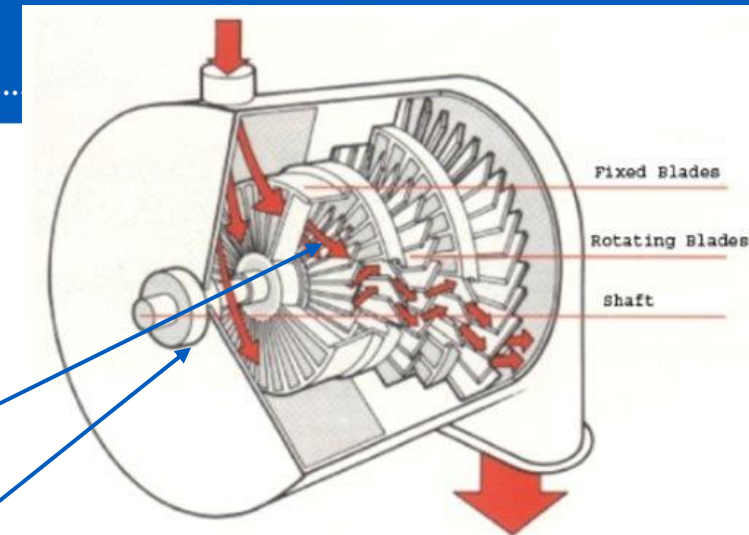
Tophat quiz



Turbines and Motors

- Turbine:

- Fixed blades and rotating blades are inside a casing.
- The rotating blades are attached to a shaft, which is then connected to a wire that rotates inside a magnetic field.
- Steam passes through, turning the rotating blades and the shaft, generating electricity



REMEMBER THAT!

- It could save your life in the zombie apocalypse

Shut up, Rick

Me neither

I do not understand how easy it is to generate electricity with the law of induction

What's induction?

What's a law?



<https://www.rollingstone.com/tv-movies/tv-movie-news/the-walking-dead-what-the-hell-happened-to-this-show-627893/>

My side rant about zombie shows

- Normally zombies kind of sit around for a long time without eating anything
- Where do they get the energy to move?
 - Like, really, they shouldn't be able to violate the law of conservation of energy, so????



I haven't eaten in weeks but somehow I can still walk and run and take down a full grown-ass human with my super strength

To remember

Concept	Formula
Teslas are a unit of magnetic field strength	None
magnetic field of a solenoid is proportional to the current and the number of turns per unit length	$B = \frac{\mu_0 i N}{L}$
the flux of a magnetic field is the product of the magnetic field and the area	$\Phi_B = B \cdot A$
a voltage is induced from a changing magnetic flux	$\mathcal{E} = \frac{\Delta \Phi_B}{\Delta t}$

Links & References

- **Magnets** — <https://en.wikipedia.org/wiki/Magnet>
- **Spintronics** — <https://www.geeksforgeeks.org/physics/spintronics/>
- **Ising model** — https://en.wikipedia.org/wiki/Ising_model
- **Magnet orientation** — <https://sdmmagnets.com/orientation-directions/>
- **Electromagnets** — <https://en.wikipedia.org/wiki/Electromagnet>
- **Solenoid coils** — <https://www.iqsdirectory.com/articles/electric-coil/solenoid-coils.html>
- **Magnetic flux** — https://en.wikipedia.org/wiki/Magnetic_flux
- **Faraday's law** — https://en.wikipedia.org/wiki/Faraday's_law_of_induction
- **Turbines** — <https://www.turbinesinfo.com/steam-turbines/>
- **Walking Dead review** — <https://www.rollingstone.com/tv-movies/tv-movie-news/the-walking-dead-what-the-hell-happened-to-this-show-627893/>
- **Zombie types** — <https://screenrant.com/every-walking-dead-zombie-type-explained-roamer-lurker-variant/>